FLATNESS AT A POINT CAN BE CHECKED AT THE ASSOCIATED POINTS OF ITS FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: an etale chart reduction with a comparison of associated points and of specializations, and a decentness argument for the converse direction

Editorial note. The label theorem-check-flatness-at-associated-points names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own theorem environment in the chapter source.

Allowed standard prerequisites (editorial).

- The associated points of the fibres of a finite type quasicoherent module, and their behaviour under etale base change
- Etale charts of a morphism and continuity of the induced map on underlying topological spaces, so that specializations can be compared
- The notion of a decent algebraic space and of a decent morphism, the representation of a point of a decent space by a quasi-compact monomorphism, and the lifting of specializations along a decent morphism
- The corresponding criterion in the case of schemes

Source reference. Original definition 03KZ: sources/stacks-chow/tags.txt, source lines 4695–4695. Prerequisite pointer invoked at proof line(s) 878; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 03I8: sources/stacks-chow/tags.txt, source lines 4594–4594. Prerequisite pointer invoked at proof line(s) 874; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03IL: sources/stacks-chow/tags.txt, source lines 4607–4607. Prerequisite pointer invoked at proof line(s) 884; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0ABX: sources/stacks-chow/tags.txt, source lines 13035–13035. Prerequisite pointer invoked at proof line(s) 880; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AC8: sources/stacks-chow/tags.txt, source lines 13046–13046. Prerequisite pointer invoked at proof line(s) 876; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original theorem 05UA: sources/stacks-chow/tags.txt, source lines 7509–7509. Prerequisite pointer invoked at proof line(s) 867; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is locally of finite type. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module of finite type. Let $x \in |X|$ with image $y \in |Y|$. Set $F = f^{-1}(\{y\}) \subset |X|$. Consider the conditions*

1. *$\mathcal{F}$ is flat at x over Y, and*
2. *for every $x' \in F \cap Ass_{X/Y}(\mathcal{F})$ which specializes to x we have that $\mathcal{F}$ is flat at x' over Y.*

Then we always have (2) $\Rightarrow$ (1). If X and Y are decent, then (1) $\Rightarrow$ (2).

Proof. Assume (2). Choose a scheme V and a surjective étale morphism $V \to Y$. Choose a scheme U and a surjective étale morphism $U \to V \times_Y X$. Choose a point $u \in U$ mapping to x. Let $v \in V$ be the image of u. We will deduce the result from the corresponding result for $\mathcal{F}|_U = (U \to X)^*\mathcal{F}$ and the point u. U_v. This works because $\mathrm{Ass}_{U/V}(\mathcal{F}|_U) \cap |U_v|$ is equal to $\mathrm{Ass}_{U_v}(\mathcal{F}|_{U_v})$ and equal to the inverse image of $F \cap \mathrm{Ass}_{X/Y}(\mathcal{F})$. Since the map $|U_v| \to F$ is continuous we see that specializations in $|U_v|$ map to specializations in F, hence condition (2) is inherited by $U \to V$, $\mathcal{F}|_U$, and the point u. Thus More on Flatness, Theorem 05UA applies and we conclude that (1) holds.

If Y is decent, then we can represent y by a quasi-compact monomorphism $\mathrm{Spec}(k) \to Y$ (by definition of decent spaces, see Decent Spaces, Definition 03I8). Then $F = |X_k|$, see Decent Spaces, Lemma 0AC8. If in addition X is decent (or more generally if f is decent, see Decent Spaces, Definition 03KZ and Decent Spaces, Lemma 0ABX), then X_y is a decent space too. Furthermore, specializations in F can be lifted to specializations in $U_v \to X_y$, see Decent Spaces, Lemma 03IL. Having said this it is clear that the reverse implication holds, because it holds in the case of schemes. □